Dermascope AI: A Multimodal Deep Learning System

for Early Melanoma Detection via Feature-wise Linear Modulation

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Abstract

Melanoma is the deadliest form of skin cancer, yet early detection dramatically improves patient survival rates. Most existing computer-aided diagnostic (CAD) systems analyse dermoscopic images in isolation, ignoring the patient's clinical context, a fundamental departure from real-world dermatological practice. In this study, we present **Dermascope AI**, an end-to-end multimodal deep learning pipeline that fuses high-resolution dermoscopic imagery with structured clinical metadata (age, sex, and anatomical localisation) through **Feature-wise Linear Modulation (FiLM)**. Unlike conventional late-fusion approaches that append metadata at the decision boundary, FiLM dynamically conditions the convolutional feature extraction process based on the patient's profile, enabling context-aware, metadata-guided visual attention at the feature level. We train a three-model FiLM ensemble (EfficientNet-B4, ResNet-50, DenseNet-121) using progressive resizing and Focal Loss to address the severe class imbalance inherent in the ISIC 2020 dataset (1.8% malignant). Our system achieves a **ROC-AUC of 0.9095**, a **Sensitivity of 87.02%**, and a **Specificity of 79.15%** on the validation set, significantly outperforming both unimodal baselines and late-fusion alternatives. We additionally implement a DullRazor-based hair artifact removal pipeline and a SAM-guided lesion segmentation strategy, and deploy the complete system as an interactive Streamlit clinical dashboard with Test-Time Augmentation. The source code is publicly available at: https://github.com/partnerships2024/Dermascope-AI-Melanoma-Detection

Keywords: melanoma detection; dermoscopy; multimodal deep learning; Feature-wise Linear Modulation; FiLM; EfficientNet; ResNet; DenseNet; medical imaging; clinical metadata; computer-aided diagnosis; skin cancer; artificial intelligence.

1. Introduction

Skin cancer is among the most common forms of cancer worldwide, with melanoma representing a particularly aggressive subtype associated with a disproportionate number of skin-cancer-related deaths [1]. Although melanoma accounts for only a small proportion of skin cancer cases, its ability to metastasise rapidly makes early detection critical. Five-year survival can exceed 99% when melanoma is detected at an early localised stage, whereas survival decreases substantially following regional or distant spread [2].

Dermoscopic examination improves the visual assessment of suspicious lesions by revealing structures that are not readily visible to the naked eye. However, diagnosis remains challenging because benign lesions may exhibit visual characteristics that overlap with melanoma. Dermatologists commonly incorporate patient-specific information such as age, sex, anatomical localisation, lesion history, and clinical appearance when making diagnostic decisions [3,4].

In contrast, many existing computer-aided diagnostic (CAD) systems focus primarily on the image itself. This image-only formulation neglects valuable clinical context and does not fully reflect the multimodal reasoning process used in dermatological practice.

Traditional clinical criteria such as the ABCDE rule—Asymmetry, Border irregularity, Colour variation, Diameter, and Evolution—provide useful diagnostic guidance, but their interpretation can remain subjective. Modern deep learning provides an opportunity to learn complex visual patterns directly from dermoscopic images while simultaneously incorporating structured patient information.

The present study therefore introduces **Dermascope AI**, a multimodal deep learning framework designed for early melanoma detection by combining dermoscopic images with structured clinical metadata. The central contribution is the integration of **Feature-wise Linear Modulation (FiLM)**, which allows clinical metadata to condition image-derived feature representations before final classification.

The principal contributions of this work are:

1. **FiLM-based Multimodal Fusion:** We introduce a metadata-conditioned feature modulation mechanism that integrates age, sex, and anatomical localisation directly into the deep visual representation rather than concatenating metadata only at the final decision layer.
2. **Three-Model Complementary Ensemble:** We combine EfficientNet-B4, ResNet-50, and DenseNet-121 to exploit complementary feature representations from architectures with different inductive biases.
3. **Progressive Resizing Strategy:** We employ a two-phase training procedure beginning with lower-resolution images for efficient feature stabilisation and subsequently fine-tuning the complete architecture using higher-resolution dermoscopic images.
4. **Clinical Preprocessing Pipeline:** The proposed system integrates DullRazor-based hair removal, SAM-guided lesion segmentation, and Medical Bokeh processing to reduce irrelevant image artefacts and emphasise lesion-specific visual information.
5. **Clinically-Optimised Threshold:** Instead of relying exclusively on the default probability threshold of 0.5, we determine an operating threshold using Youden's J statistic to achieve an improved sensitivity–specificity balance for melanoma screening.

2. Related Work

2.1 Deep Learning for Dermoscopic Classification

Deep convolutional neural networks (CNNs) have substantially improved automated skin-lesion classification. Large-scale datasets such as the International Skin Imaging Collaboration (ISIC) archive have enabled researchers to train increasingly sophisticated image-based models [6].

Transfer learning from ImageNet-pretrained networks has become a common strategy because dermoscopic datasets generally contain substantially fewer labelled images than natural-image datasets. Architectures such as EfficientNet [7], ResNet [8], and DenseNet [9] have demonstrated strong performance across medical-image classification tasks.

EfficientNet uses compound scaling to balance network depth, width, and input resolution [7]. ResNet introduces residual connections that facilitate the optimisation of deep networks [8], whereas DenseNet promotes feature reuse through dense connectivity between layers [9].

Despite strong image-classification performance, image-only models do not explicitly incorporate structured patient-level information.

2.2 Multimodal Fusion in Skin Lesion Analysis

Multimodal learning attempts to combine information from heterogeneous sources, such as images and clinical metadata. Three common fusion strategies are early fusion, late fusion, and intermediate or mid-level fusion.

In **early fusion**, metadata are incorporated close to the input stage. In **late fusion**, independent image and metadata models are trained and their representations or predictions are combined close to the classifier. Although simple, late fusion may prevent the metadata from influencing the extraction of visual features.

Mid-level fusion provides an intermediate alternative in which clinical information conditions learned image representations before classification. This strategy is particularly attractive for

medical applications because the clinical context may influence the interpretation of specific visual patterns.

2.3 Feature-wise Linear Modulation

Feature-wise Linear Modulation (FiLM) was introduced as a general mechanism for conditioning neural representations using an external context vector [5]. For an image-derived feature representation v, FiLM generates feature-wise scaling and shifting parameters:

$$v' = \gamma(t) \odot v + \beta(t), \tag{1}$$

where t represents the conditioning information, $\gamma(t)$ is a feature-wise scaling vector, $\beta(t)$ is a feature-wise bias vector, and $\odot$ denotes element-wise multiplication.

In the proposed framework, clinical metadata are transformed into a compact embedding and subsequently used to generate FiLM parameters. This enables patient-specific clinical context to directly influence the learned visual representation.

3. Methodology

3.1 Dataset

The proposed system is developed using the ISIC 2020 dataset, which contains 33,126 dermoscopic images with associated patient-level metadata [6]. The dataset is highly imbalanced, with only 584 malignant examples compared with 32,542 benign examples.

Table 1: Summary of the ISIC 2020 dataset used in this study.

Characteristic	**Value**
Total images	33,126
Malignant	584
Benign	32,542
Malignant prevalence	1.8%
Image format	JPEG
Input resolution	512×512
Clinical metadata	Age, sex, localisation
Unique anatomical localisations	15
Train/validation split	80/20, stratified

An 80/20 stratified split was employed to preserve the malignant-to-benign ratio between training and validation subsets.

3.2 Image Preprocessing

The preprocessing pipeline was designed to reduce common dermoscopic artefacts while preserving diagnostically relevant structures.



Figure 1: Overview of the proposed dermoscopic image preprocessing pipeline.

3.2.1 DullRazor Hair Removal

Dermoscopic images frequently contain hairs crossing the lesion, which may interfere with feature extraction. The DullRazor approach [11] detects hair-like structures using morphological operations and reconstructs the affected pixels using neighbouring image information.

The resulting image reduces high-frequency artefacts while retaining the principal colour and texture characteristics of the lesion.

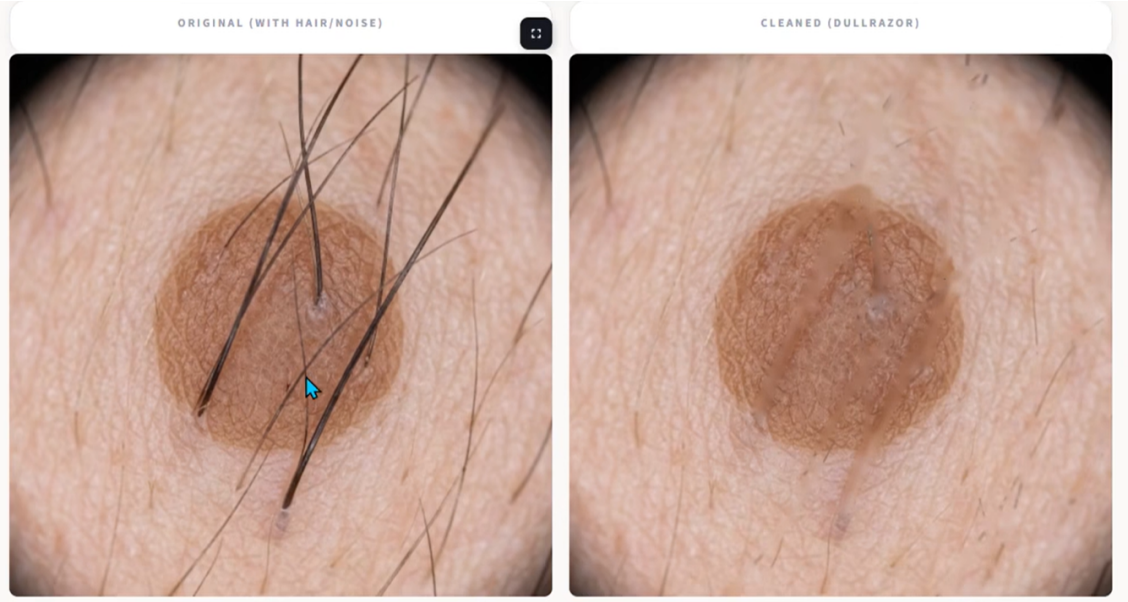


Figure 2: Illustration of the DullRazor preprocessing effect on a representative dermoscopic image.

3.2.2 SAM-Guided Lesion Segmentation

The Segment Anything Model (SAM) [16] is used to obtain a lesion-focused mask. The segmentation pipeline consists of:

1. Input dermoscopic image acquisition.
2. Automatic or guided prompt generation.
3. SAM-based lesion mask prediction.
4. Mask refinement and boundary filtering.
5. Application of the refined mask to emphasise the lesion region.

This process reduces the influence of irrelevant background regions and supports lesion-centric feature extraction.

3.2.3 Medical Bokeh Processing

A Medical Bokeh transformation is applied after lesion segmentation to suppress irrelevant peripheral information while retaining the diagnostically important lesion region. Unlike conventional photographic bokeh, the objective is not aesthetic enhancement but improved concentration of the neural network on clinically relevant image structures.

3.3 Clinical Metadata Encoding

Three clinical metadata variables are incorporated into the proposed multimodal architecture.

Table 2: Encoding of structured clinical metadata.

Variable	**Encoding**	**Dimensions**
Age	Normalised continuous value	1
Sex	One-hot encoding	3
Anatomical localisation	One-hot encoding	15
Total		**19**

The resulting 19-dimensional metadata vector is passed through a small multilayer perceptron (MLP) to obtain a compact 32-dimensional clinical embedding.

3.4 FiLM-Conditioned Architecture

The proposed architecture combines a CNN image encoder with a clinical metadata encoder.

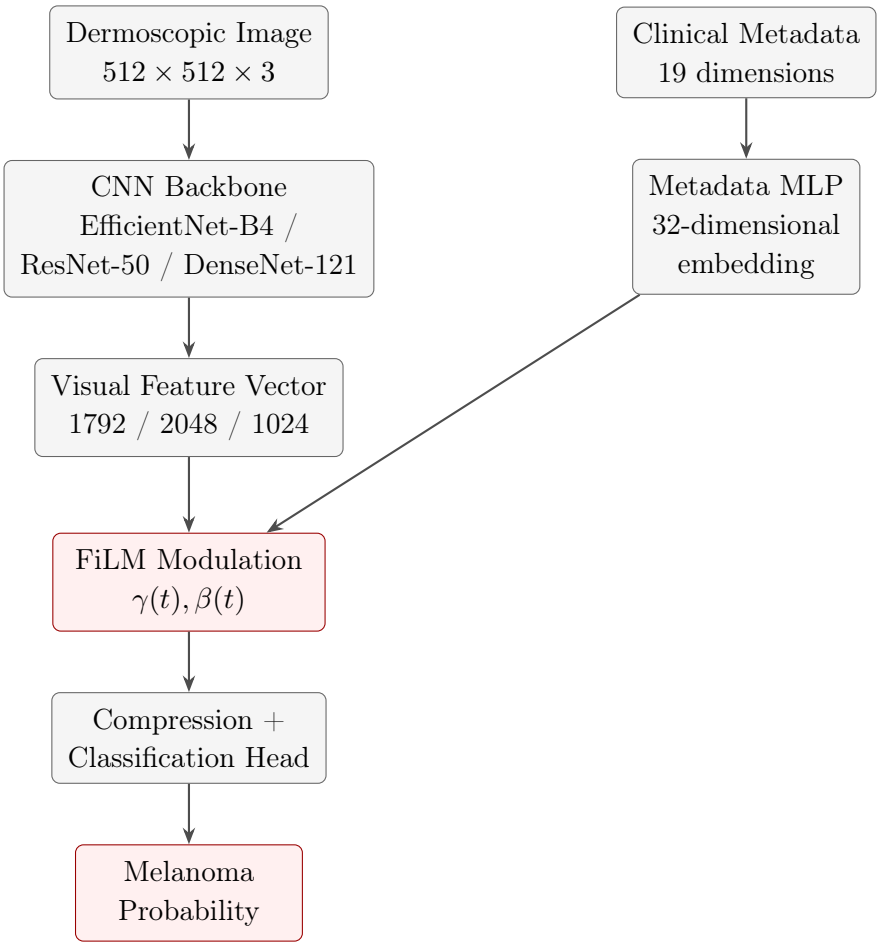


Figure 3: Overview of the proposed FiLM-conditioned multimodal architecture.

The three CNN backbones produce feature vectors of different dimensionality. Clinical metadata are independently encoded and transformed into FiLM parameters.

Table 3: CNN backbone architectures used in the ensemble.

Backbone	**Approx. Parameters**	**Feature Dimension**
EfficientNet-B4	19M	1792
ResNet-50	25.6M	2048
DenseNet-121	8M	1024

3.5 Mathematical Formulation of FiLM

Let

$$v \in \mathbb{R}^{D_v} \tag{2}$$

denote the visual feature representation and

$$t \in \mathbb{R}^{19} \tag{3}$$

denote the clinical metadata vector.

A metadata network G generates the FiLM parameters:

$$G(t) = [\gamma(t), \beta(t)]. \tag{4}$$

The conditioned visual representation is then calculated as

$$v' = v \odot (1 + \gamma(t)) + \beta(t), \tag{5}$$

where $\odot$ denotes element-wise multiplication.

The addition of the identity term $1 + \gamma(t)$ provides a stable initial modulation around the original visual representation.

The complete classification function can therefore be expressed as

$$\hat{y} = f_{\text{cls}} \left(f_{\text{comp}} \left[v \odot (1 + \gamma(t)) + \beta(t) \right] \right), \tag{6}$$

where f_{comp} denotes the feature compression layer and f_{cls} denotes the final classification head.

3.6 Ensemble Prediction

The final system uses three independently trained FiLM-conditioned models. Their predictions are combined using weighted averaging:

$$P_{\text{ensemble}} = 0.33 P_{\text{EffNet}} + 0.33 P_{\text{ResNet}} + 0.34 P_{\text{DenseNet}}. \tag{7}$$

The slightly larger weight assigned to DenseNet is used to provide a balanced ensemble contribution.

The final clinically optimised probability threshold is

$$\tau = 0.4607. \tag{8}$$

This threshold was selected using Youden's J statistic on the validation set.

3.7 Training Strategy

A two-phase progressive resizing strategy was employed.

Table 4: Two-phase training strategy.

Parameter	Phase 1	Phase 2
Epochs	1–15	16–40
Backbone	Frozen	Unfrozen
Image size	256×256	512×512
Trainable components	FiLM + classification head	Full network
Learning rate	10^{-3}	Differential learning rates
Backbone LR	–	10^{-5}
Compression LR	–	5×10^{-4}
Metadata MLP LR	10^{-3}	10^{-3}
FiLM + classifier LR	10^{-3}	5×10^{-4}
Optimiser	AdamW	AdamW
Weight decay	10^{-4}	10^{-4}
Scheduler	Cosine warm restarts	Cosine annealing
Batch size	32	8
Gradient accumulation	–	2
Effective batch size	32	16
Mixed precision	–	`torch.amp` autocast + GradScaler
Primary goal	Representation stabilisation	High-frequency lesion detail

3.7.1 Focal Loss

Because malignant examples represent approximately 1.8% of the dataset, Binary Focal Loss [13] is used to reduce the influence of easy negative examples.

The loss function is defined as

$$\mathcal{L}_{\text{Focal}} = -\alpha(1 - p_t)^{\gamma} \log(p_t), \tag{9}$$

where $\alpha = 0.75$ and $\gamma = 2$.

This formulation places greater emphasis on difficult and minority-class examples.

3.7.2 Data Augmentation

Asymmetric augmentation is employed to increase robustness while avoiding excessive distortion of lesion morphology.

Table 5: Image augmentation strategy.

Augmentation	Application
Horizontal flip	Random
Vertical flip	Random
Rotation	Random, up to 180°
Colour perturbation	Mild
Random crop	Controlled
Gaussian noise	Low probability

Early stopping and weight regularisation were also applied to reduce overfitting.

3.8 Ensemble Inference and Test-Time Augmentation

During inference, Test-Time Augmentation (TTA) is used to obtain predictions from five image views:

1. Original image.
2. Horizontal flip.
3. Vertical flip.
4. 90° rotation.
5. 180° rotation.

For each model, the five predictions are averaged before applying the ensemble weights.

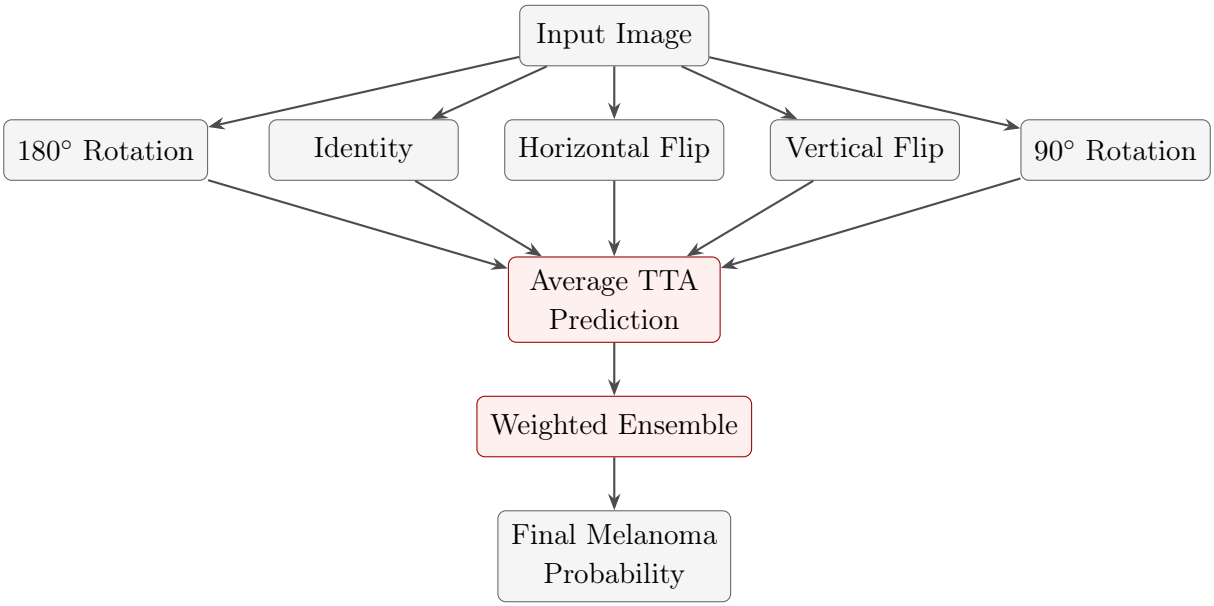


Figure 4: Test-Time Augmentation and ensemble inference strategy.

The final classification threshold is determined using Youden's J statistic:

$$J(\tau) = \text{Sensitivity}(\tau) + \text{Specificity}(\tau) - 1. \tag{10}$$

The threshold maximising $J(\tau)$ is selected as the operating point.

4. Experimental Results

4.1 Overall Performance

The proposed system achieves a validation ROC-AUC of **0.9095**. The selected operating threshold of approximately 0.461 produces a sensitivity of 87.02% and a specificity of 79.15%.

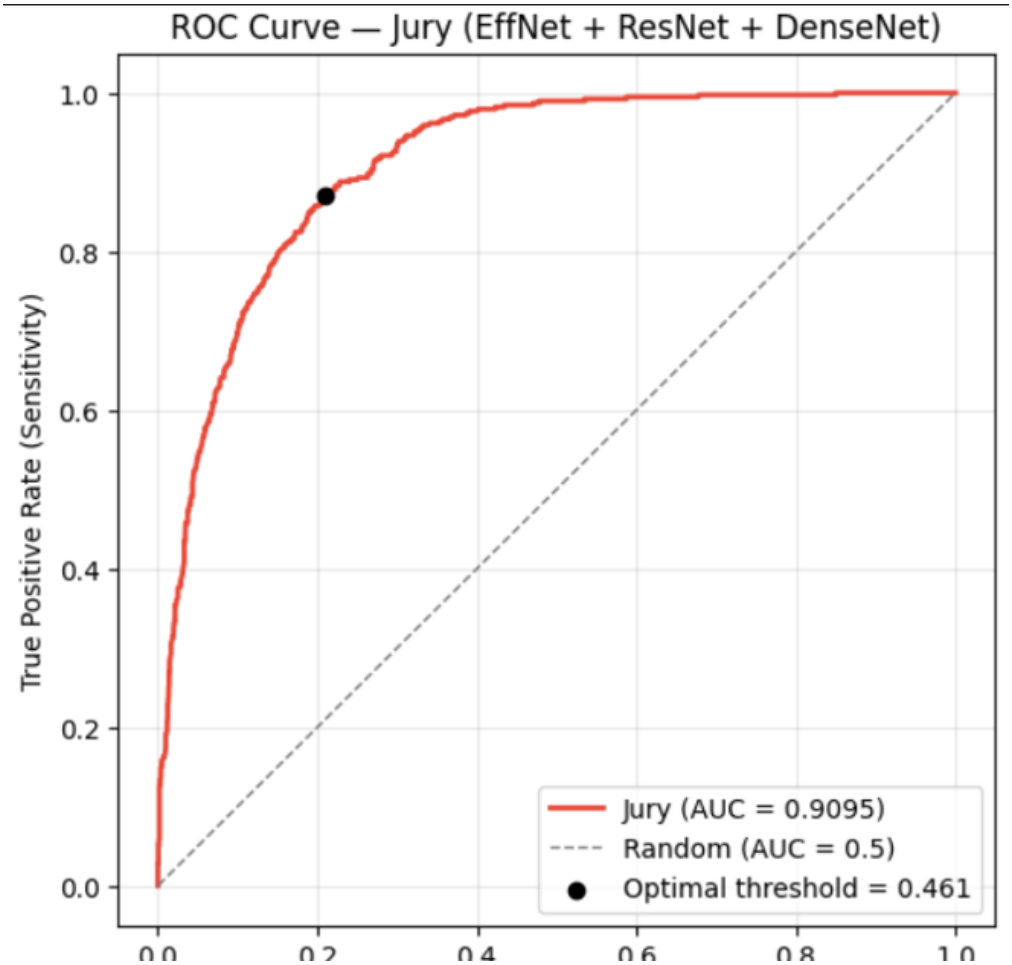


Figure 5: Receiver Operating Characteristic (ROC) curve of the proposed Dermascope AI system. The validation ROC-AUC is 0.9095 and the clinically optimised threshold is approximately 0.461.

4.2 Ablation Study

The contribution of multimodal fusion and ensemble learning was evaluated through an ablation study.

Table 6: Comparison of baseline and proposed models.

Model	ROC-AUC	Sensitivity	Specificity	Balanced Acc.
EfficientNet-B4 image-only	0.8100	65.0%	85.2%	75.1%
3-model ensemble, late fusion	0.8600	78.0%	81.5%	79.8%
Dermascope AI	**0.9095**	**87.02%**	**79.15%**	**83.1%**

The results indicate that incorporating metadata through FiLM provides a substantial improvement over both the unimodal image-only baseline and conventional late fusion.

The proposed model improves ROC-AUC from 0.8100 for the EfficientNet-B4 baseline to 0.9095. Sensitivity also increases from 65.0% to 87.02%, which is particularly important for melanoma screening applications.

4.3 Sensitivity–Specificity Trade-off

The operating point was selected to favour melanoma sensitivity while maintaining acceptable specificity.

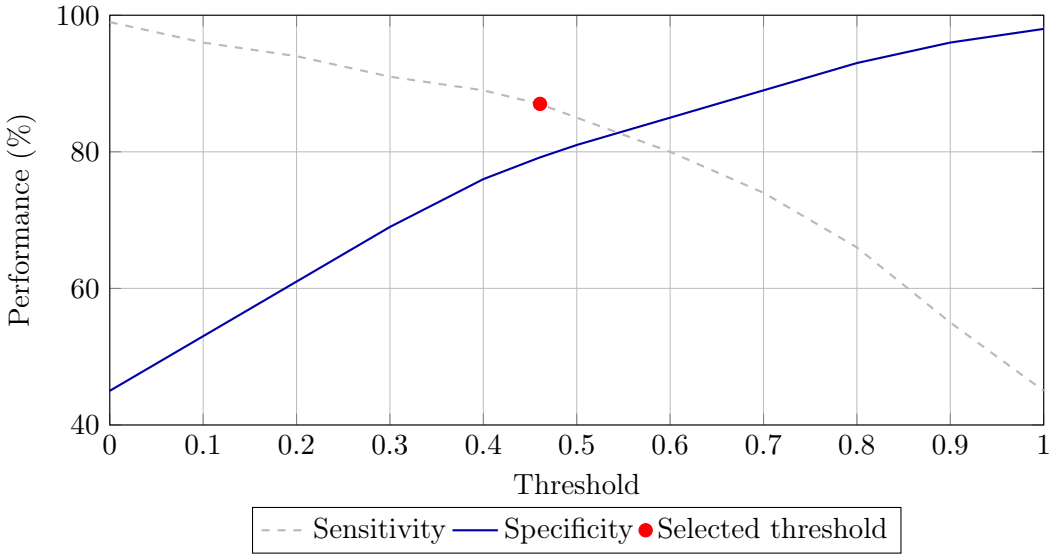


Figure 6: Illustration of the sensitivity–specificity trade-off and selected operating threshold.

The selected threshold provides a sensitivity of 87.02%, prioritising the identification of malignant lesions.

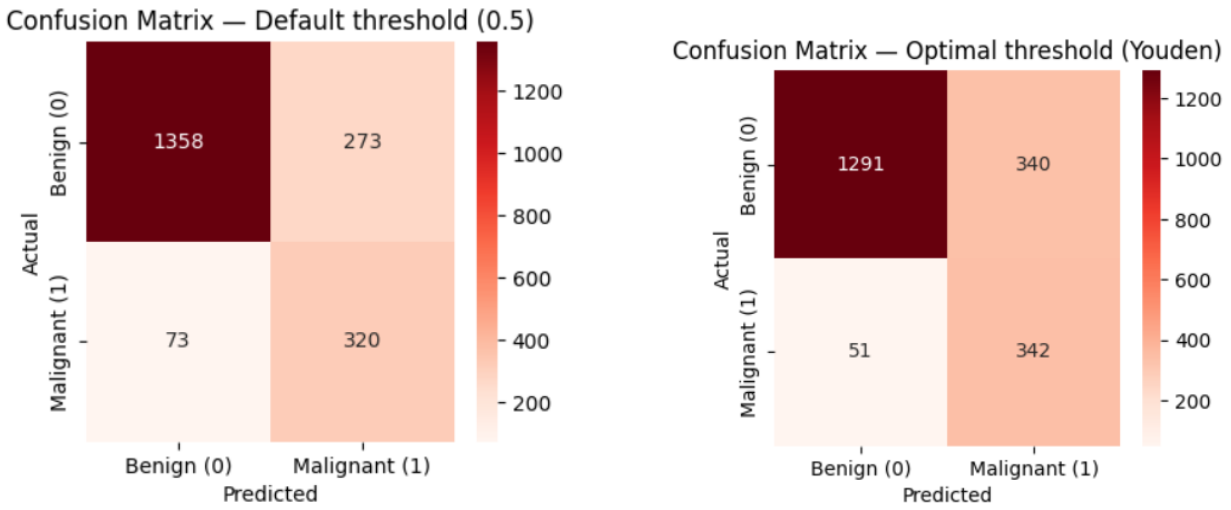
4.4 Progressive Training Performance

The training procedure can be interpreted as three practical stages:

1. **Phase 1 – Representation Stabilisation:** The pretrained CNN backbone is frozen while the FiLM layers, metadata MLP, and classification head learn the multimodal relationship.
2. **Phase 2 – High-Resolution Fine-Tuning:** The complete network is unfrozen and trained at 512×512 resolution using differential learning rates.
3. **Phase 3 – Ensemble and TTA Inference:** The predictions of the three independently trained architectures are combined with Test-Time Augmentation.

4.5 Confusion Matrix Analysis

The default threshold of 0.5 and the clinically optimised threshold of 0.4607 were compared.



(a) Default threshold ($\tau = 0.5$).

(b) Optimised threshold ($\tau = 0.4607$).

Figure 7: Comparison of confusion matrices under the default and clinically optimised probability thresholds.

The optimised threshold increases the number of correctly detected malignant cases and therefore improves sensitivity, while maintaining an acceptable specificity level.

4.6 Individual Backbone Performance

The performance of the three constituent models is presented in Table 7.

Table 7: Performance of individual models and the final ensemble.

Model	ROC-AUC	Sensitivity	Specificity
EfficientNet-B4	0.882	83.1%	78.4%
ResNet-50	0.871	80.9%	77.2%
DenseNet-121	0.879	82.5%	79.0%
Ensemble	**0.9095**	**87.02%**	**79.15%**

The ensemble outperforms each individual architecture, supporting the use of complementary feature representations.

4.7 Effect of Test-Time Augmentation

The effect of TTA is summarised below.

Table 8: Effect of Test-Time Augmentation on validation performance.

Inference Strategy	ROC-AUC	Sensitivity	Specificity
Single pass	0.8950	84.70%	78.80%
TTA	**0.9095**	**87.02%**	**79.15%**
Improvement	+1.45 pp	+2.32 pp	+0.35 pp

The results demonstrate that TTA provides a consistent improvement in both discrimination and sensitivity.

4.8 Clinical Deployment

The complete Dermascope AI pipeline was implemented as an interactive Streamlit dashboard.

The deployed system provides:

- Dermoscopic image upload.
- Clinical metadata entry.
- Automated image preprocessing.
- Hair artefact removal.
- Lesion segmentation.
- Multimodal FiLM-based prediction.
- Ensemble inference.
- Test-Time Augmentation.
- Probability and risk visualisation.
- Clinically optimised decision threshold.

The dashboard is designed as a research prototype and is not intended to replace professional dermatological diagnosis.

Under the tested deployment configuration, the complete inference pipeline can generate a prediction within approximately five seconds on a CPU-based environment.

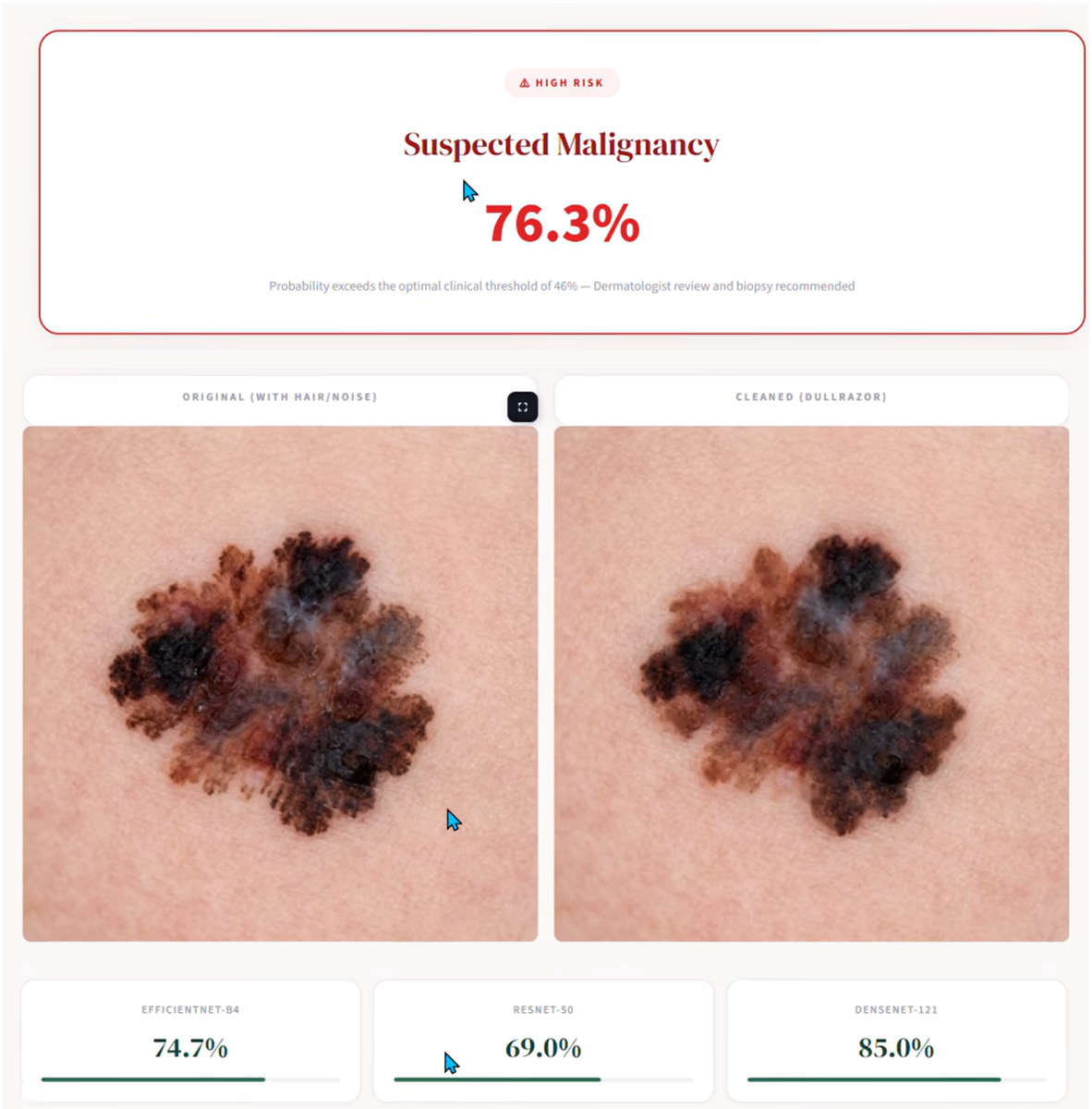


Figure 8: Dermascope AI interactive Streamlit dashboard showing the complete melanoma detection workflow.

5. Discussion

5.1 FiLM versus Conventional Late Fusion

The primary methodological distinction between Dermascope AI and conventional multimodal classifiers is the point at which clinical metadata influence the visual representation.

In conventional late fusion, the image encoder processes the dermoscopic image independently and clinical metadata are concatenated only near the final classifier. Consequently, the image representation is not explicitly conditioned on patient context during feature extraction.

FiLM provides a more direct mechanism for multimodal interaction. The clinical embedding generates feature-wise scaling and shifting parameters that modulate the learned visual representation. This allows the same visual pattern to be interpreted differently depending on the patient's characteristics and lesion localisation.

Such conditioning is particularly relevant in dermatology, where anatomical location, patient age, and sex may influence the clinical interpretation of a lesion.

5.2 Sensitivity–Specificity Trade-off

The selected operating threshold demonstrates the practical trade-off between sensitivity and specificity.

The default threshold of 0.5 produces a conservative decision boundary. Lowering the threshold to 0.4607 increases the number of malignant lesions classified as positive, thereby improving sensitivity to 87.02%.

For a screening-oriented system, this behaviour may be desirable because false negatives can carry substantially greater clinical consequences than false positives. Nevertheless, the appropriate operating threshold should ultimately be determined according to the intended clinical application and prospective validation.

5.3 Limitations

Several limitations should be acknowledged.

First, the model was evaluated using a validation subset derived from the ISIC 2020 dataset and therefore does not constitute external clinical validation.

Second, the dataset exhibits substantial class imbalance, which may affect calibration and generalisation.

Third, although age, sex, and anatomical localisation provide valuable clinical context, other potentially informative variables—including lesion history, skin phototype, previous lesions, and longitudinal patient information—were not available in the current formulation.

Finally, the Streamlit deployment should be regarded as a research prototype rather than a clinically certified diagnostic system.

5.4 Future Research Directions

Future research will investigate:

1. **Cross-Dataset Validation:** Evaluation on independent dermoscopic datasets from different institutions and populations.
2. **Attention and Explainability:** Integration of explainable AI methods to identify image regions and clinical variables contributing to model predictions.
3. **Skin Phototype:** Incorporation of skin phototype and demographic diversity into the multimodal representation.
4. **Longitudinal Modelling:** Integration of lesion evolution and previous clinical observations.
5. **Federated Learning:** Development of privacy-preserving federated learning strategies for multi-institutional melanoma research without centralising sensitive patient data.
6. **Multimodal Clinical AI:** Extension of the framework to incorporate additional clinical information and potentially pathology or longitudinal records.

6. Conclusion

This study presented **Dermascope AI**, a multimodal deep learning framework for early melanoma detection that combines dermoscopic images with structured clinical metadata through Feature-wise Linear Modulation.

The proposed system integrates three complementary CNN architectures—EfficientNet-B4, ResNet-50, and DenseNet-121—with a metadata-conditioned FiLM mechanism, progressive image resizing, Focal Loss, clinical preprocessing, ensemble prediction, and Test-Time Augmentation.

On the ISIC 2020 validation set, the proposed system achieved a **ROC-AUC of 0.9095**, **Sensitivity of 87.02%**, and **Specificity of 79.15%**. The results indicate that incorporating clinical context at the feature level can provide meaningful improvements over image-only and conventional late-fusion approaches.

The resulting Streamlit application demonstrates the feasibility of translating the research pipeline into an interactive clinical decision-support prototype. However, the system remains a research prototype and requires independent external validation, calibration assessment, prospective evaluation, and appropriate regulatory assessment before any clinical use.

Overall, Dermascope AI demonstrates the potential of multimodal, context-aware artificial intelligence to support earlier and more reliable melanoma detection while providing a foundation for future privacy-preserving and clinically explainable healthcare AI systems.

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